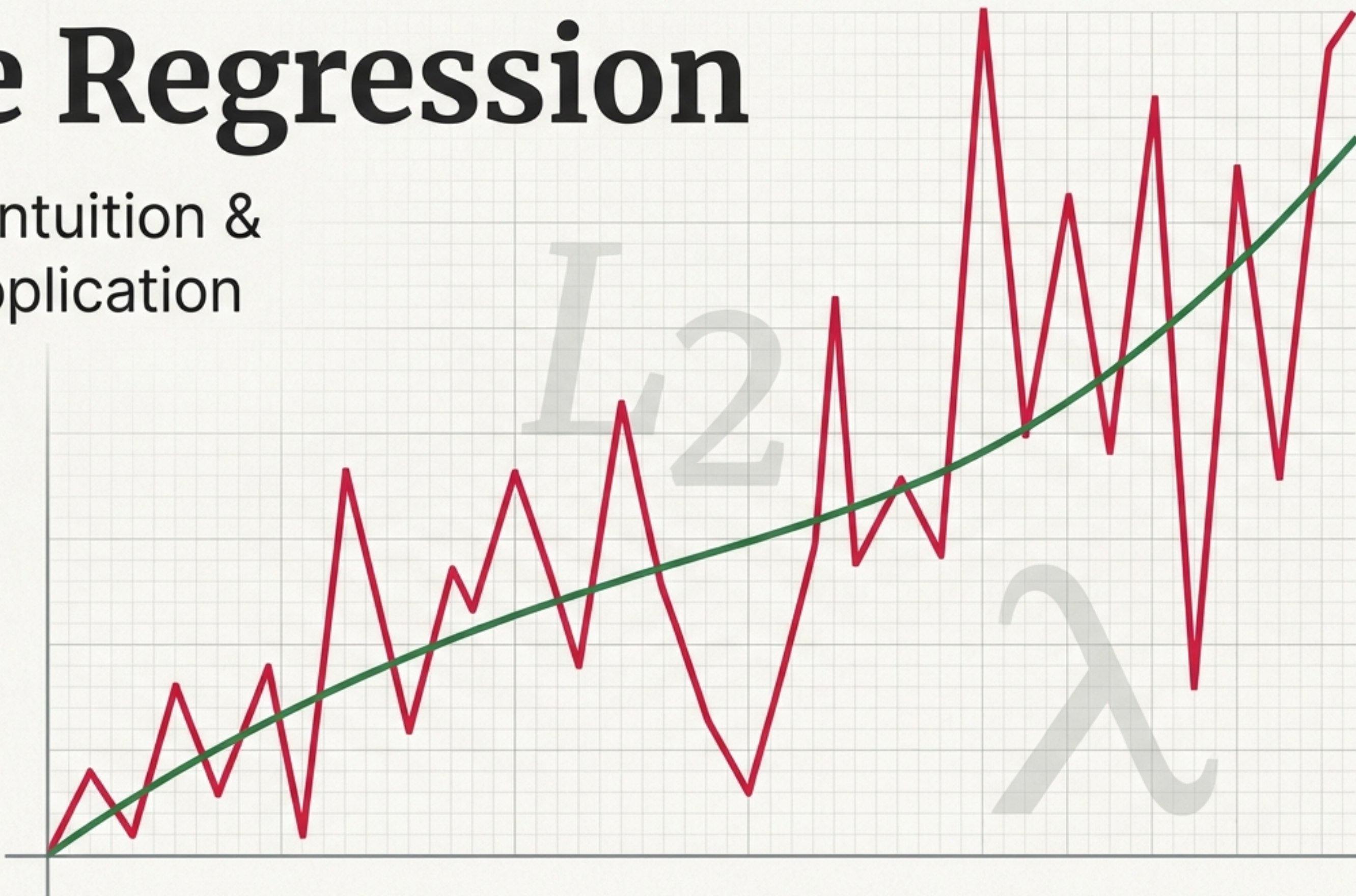
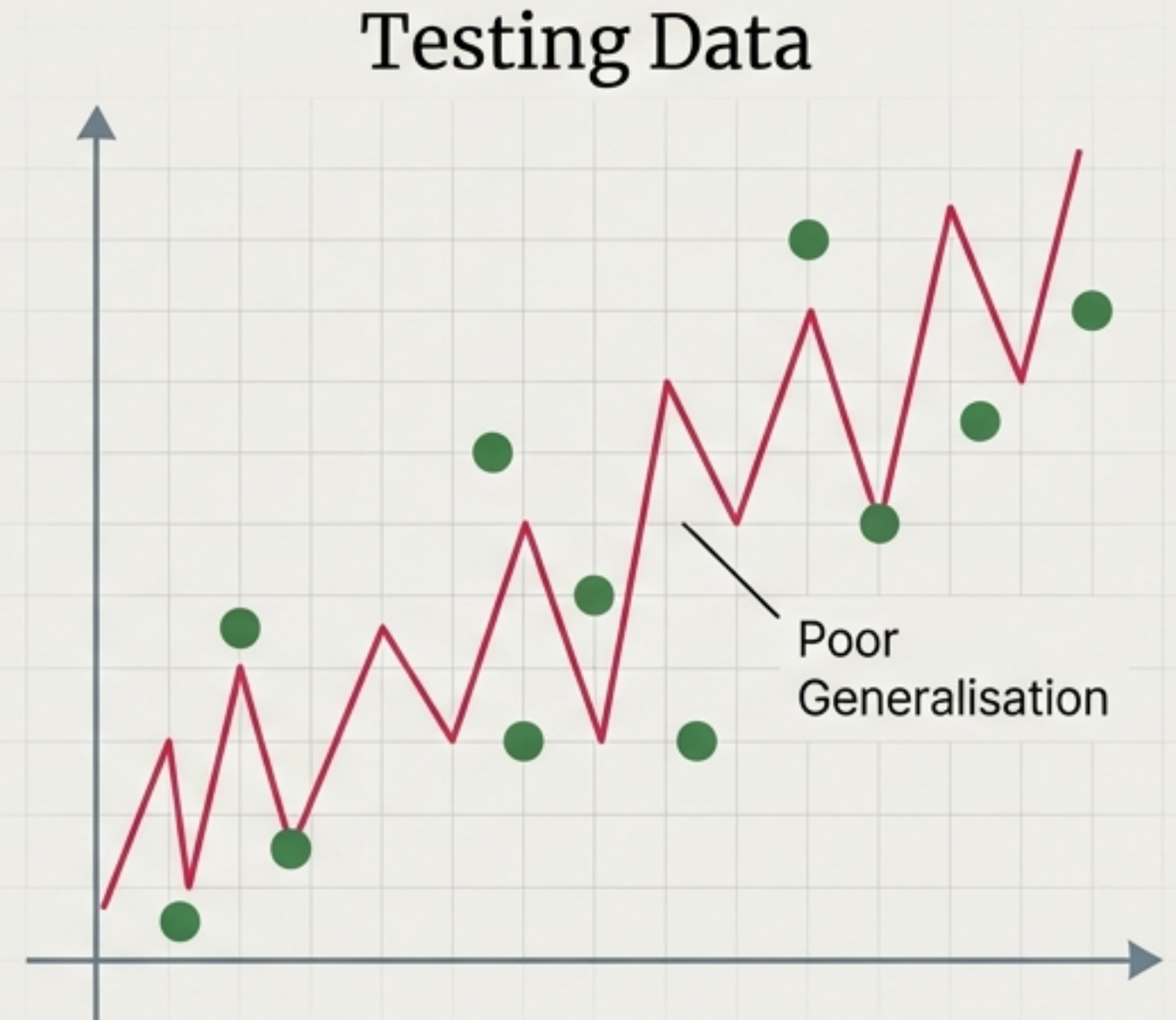
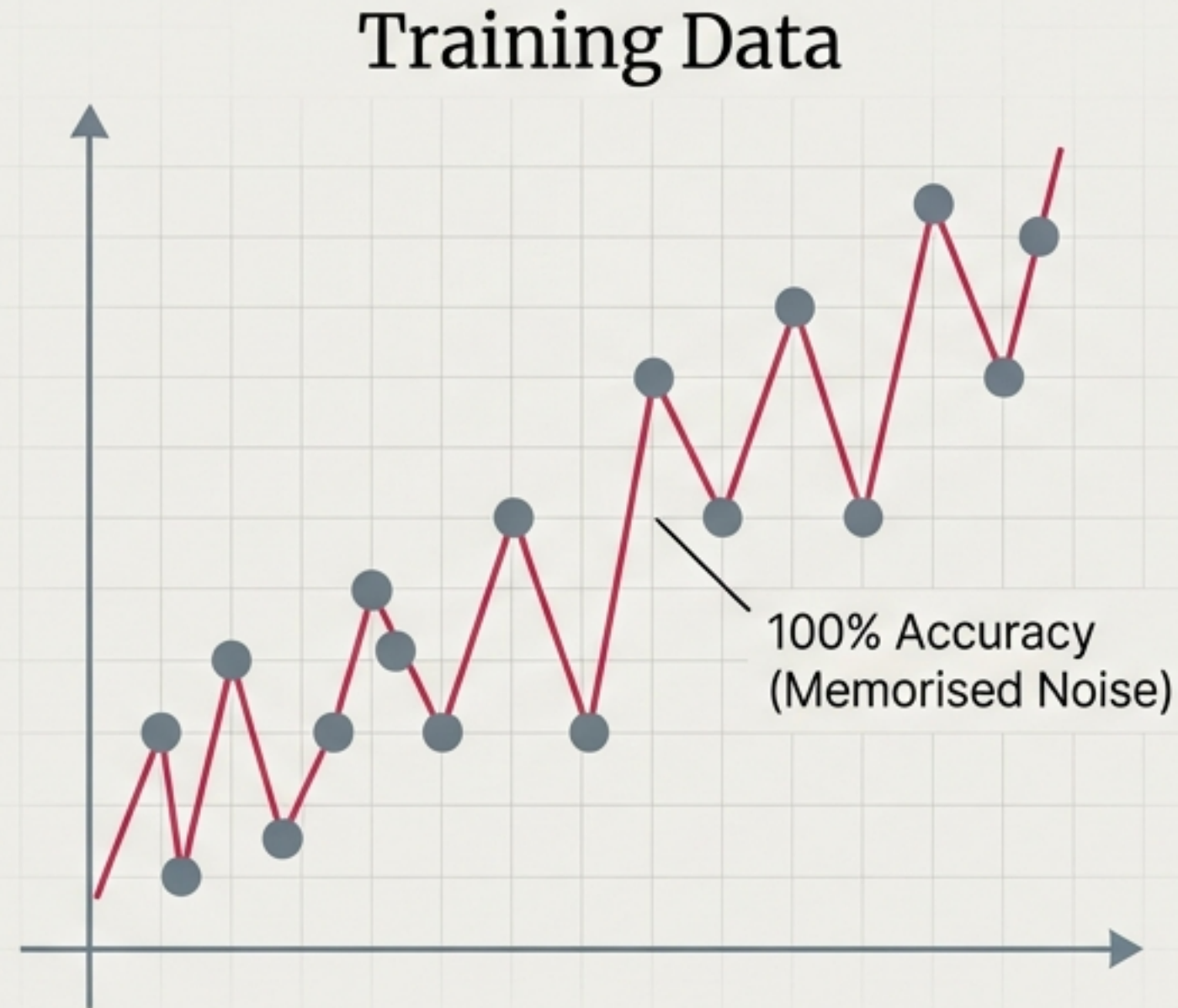


Ridge Regression

Geometric Intuition &
Practical Application



Machine learning models often memorise rather than generalise.



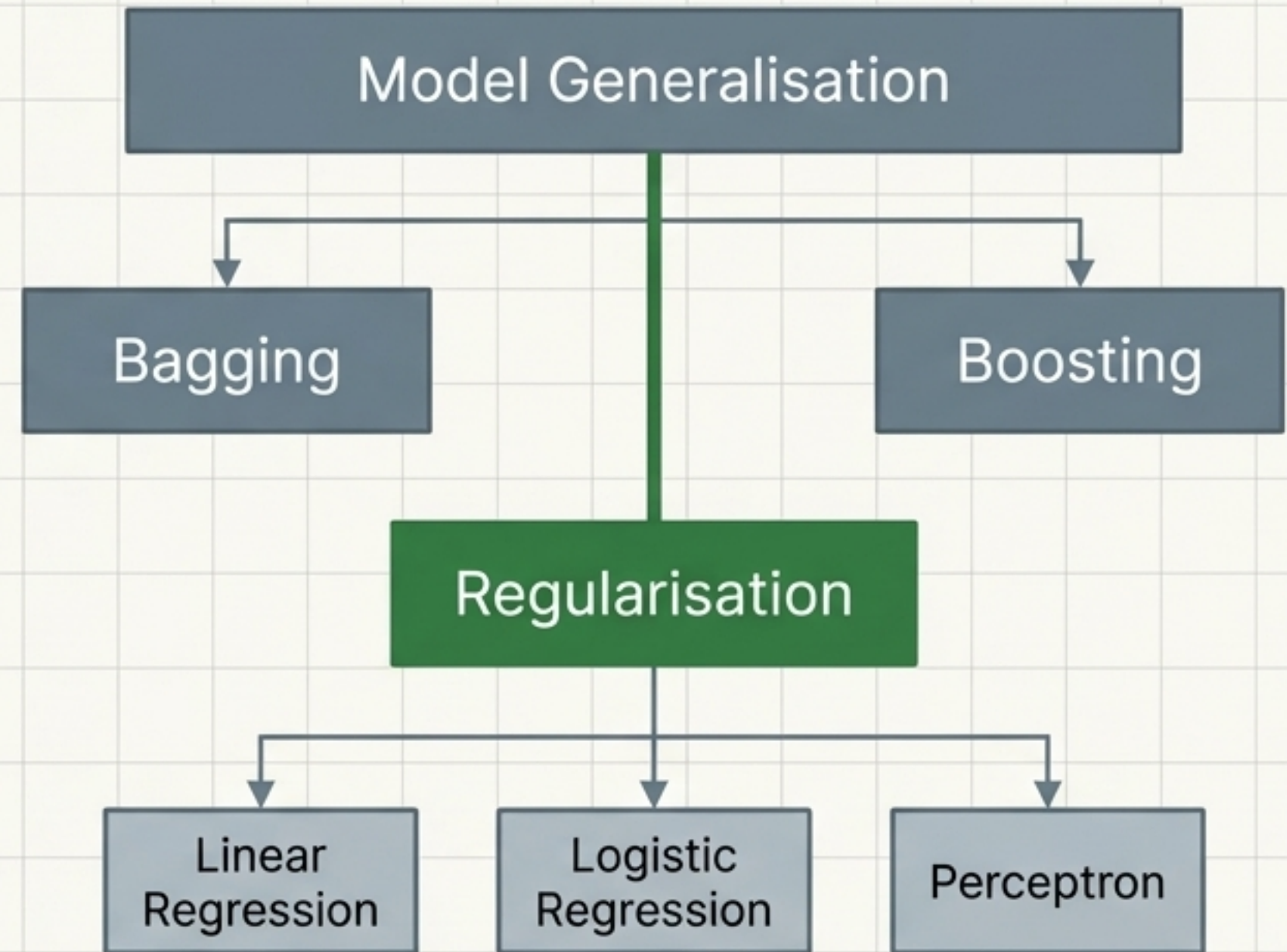
- Overfitting occurs when a model performs exceptionally well on its training data, but delivers completely different, poor results on testing data.
- The model has learned the noise, not the underlying signal.

Regularisation induces additional information to penalise model complexity

To stop models from over-learning, we must constrain them.

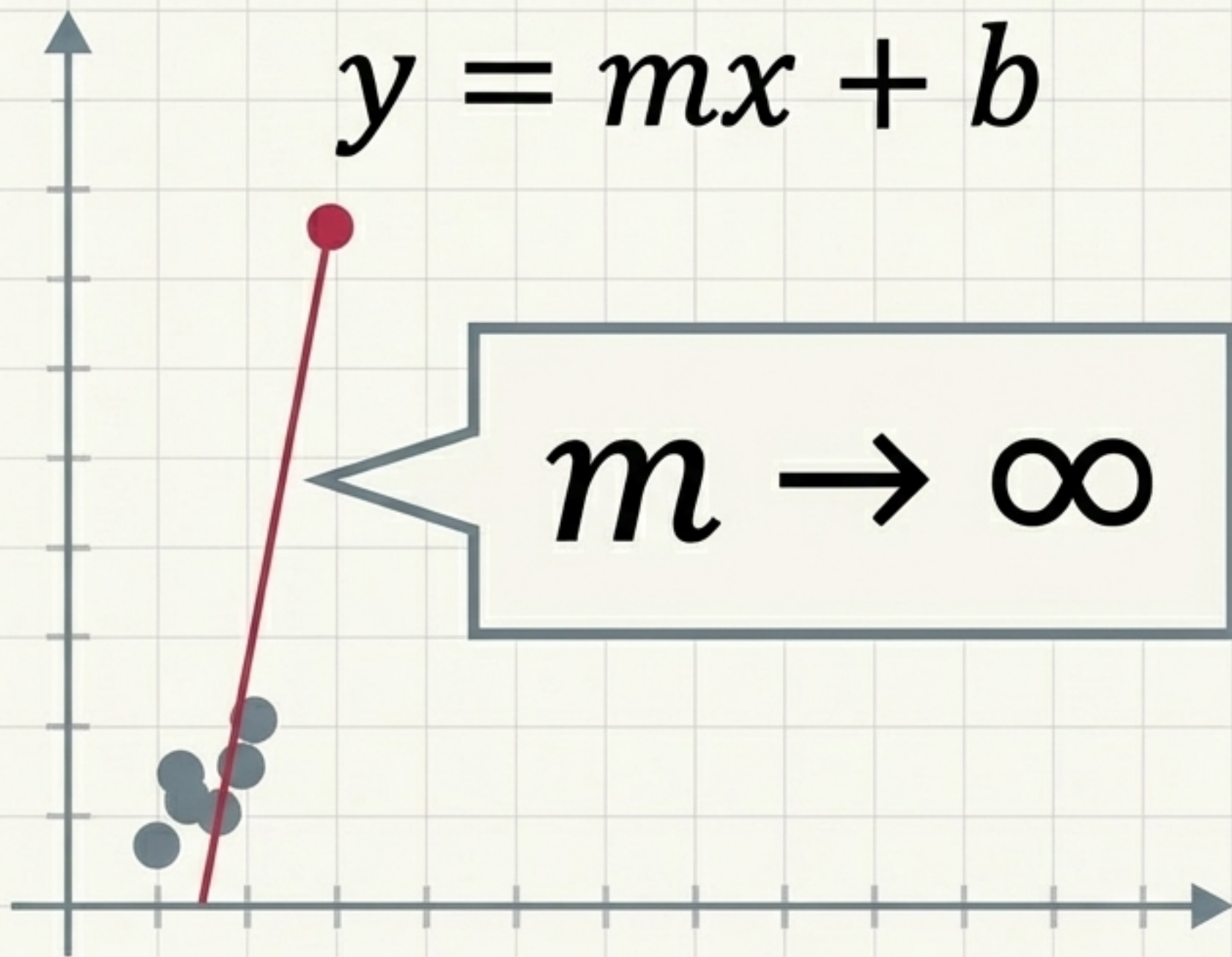
The 'Big Three' Regularisation techniques:

1. Lasso Regression (L1)
2. Ridge Regression (L2)
3. Elastic Net (A combination of Ridge and Lasso)



Standard linear regression can generate extreme slopes to force a perfect fit.

- The core objective of linear regression is to find the best fit line.
- Without constraints, the system is completely undefeated.
- To hit every single training point and minimise standard error, the slope can grow infinitely large.



Ridge regression fixes this by introducing an L2 penalty to the loss function.

$$[\text{Standard Loss Function}] + [\lambda m^2]$$

The L2 Penalty

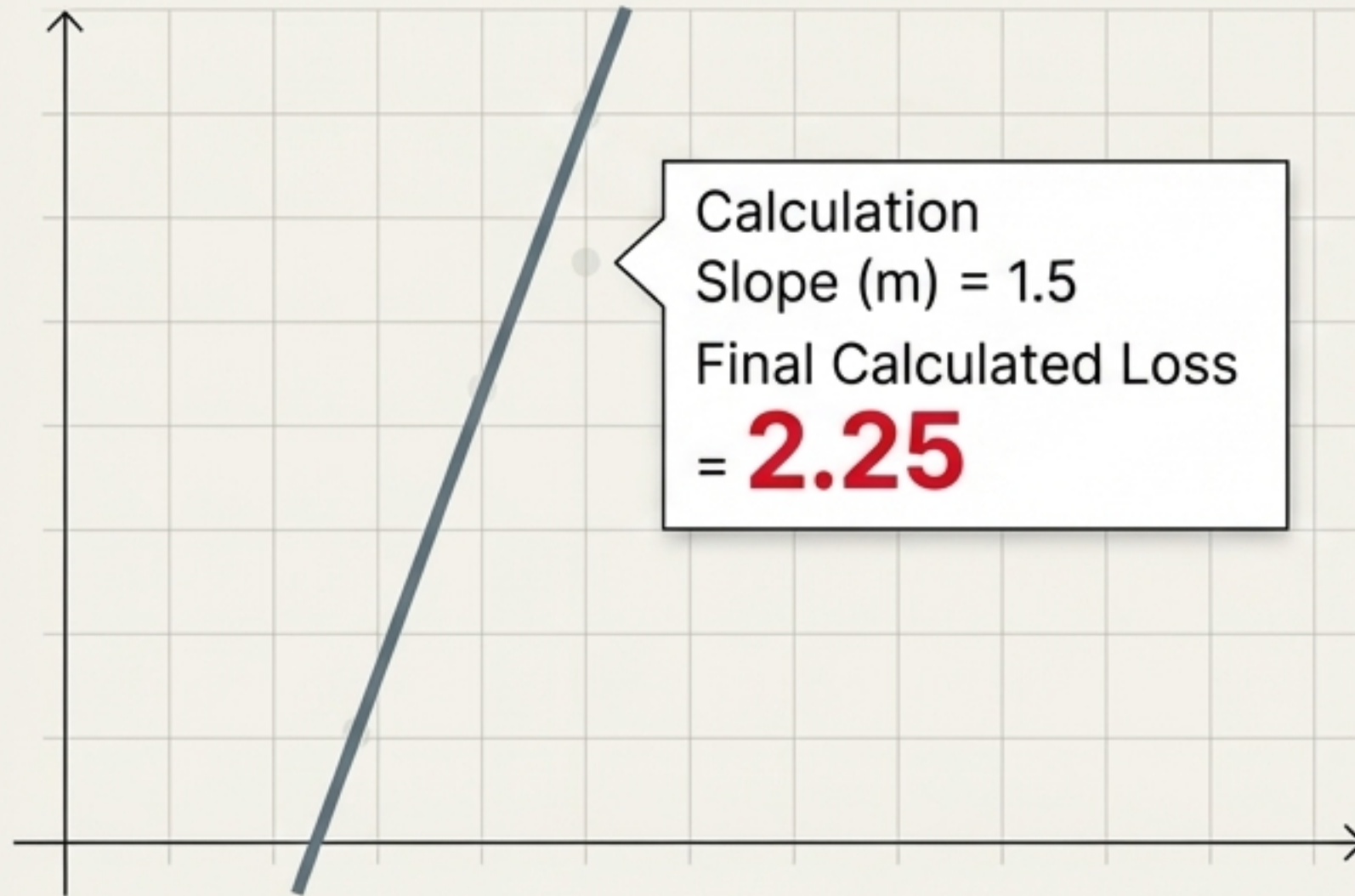


We add an extra term to the loss function calculation.

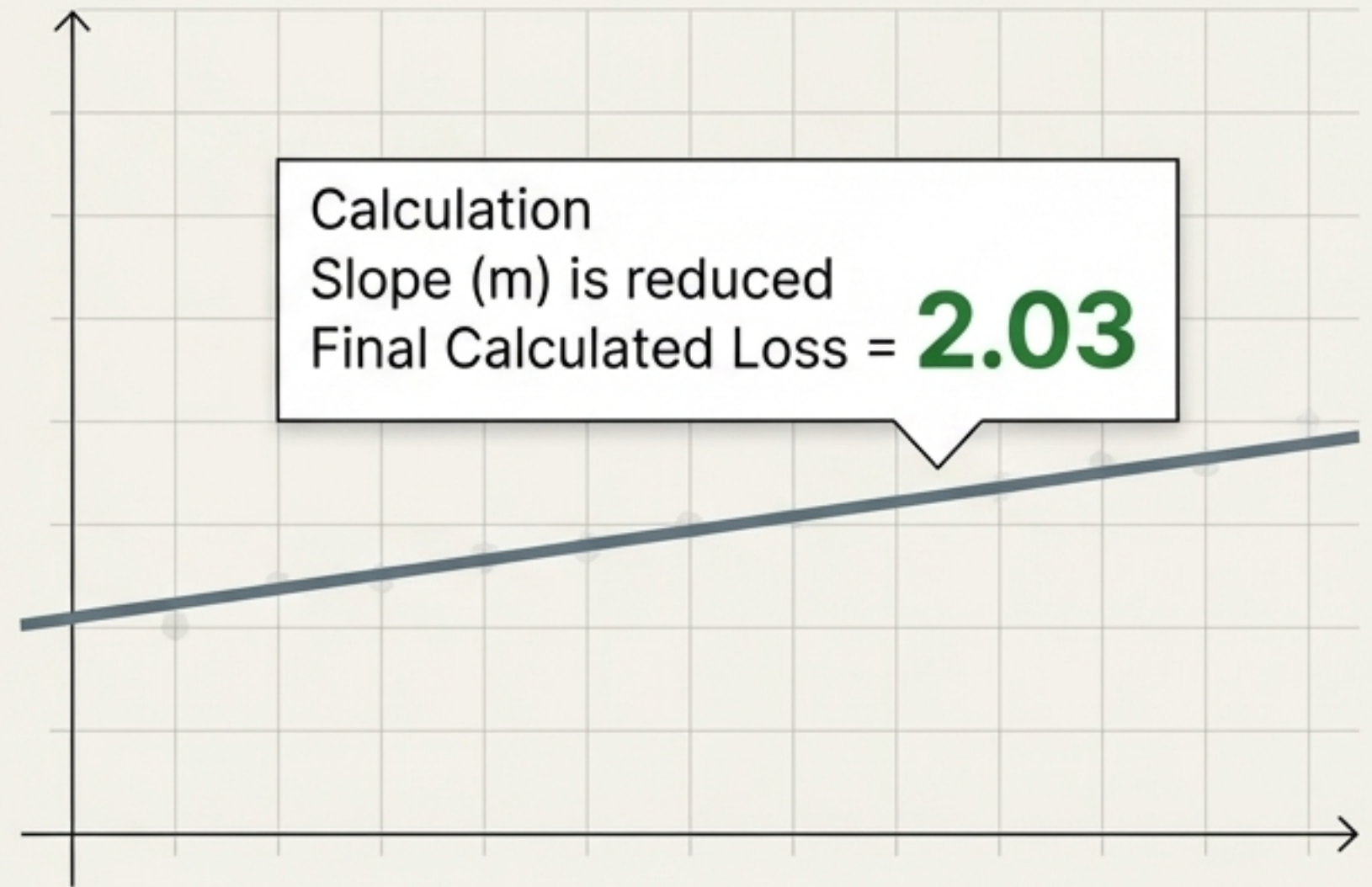
By squaring the slope (m) and multiplying it by a tuning parameter (λ), the model mathematically punishes extreme slopes.

Comparing loss calculations reveals how Ridge selects simpler models

Line 1 (Unconstrained)

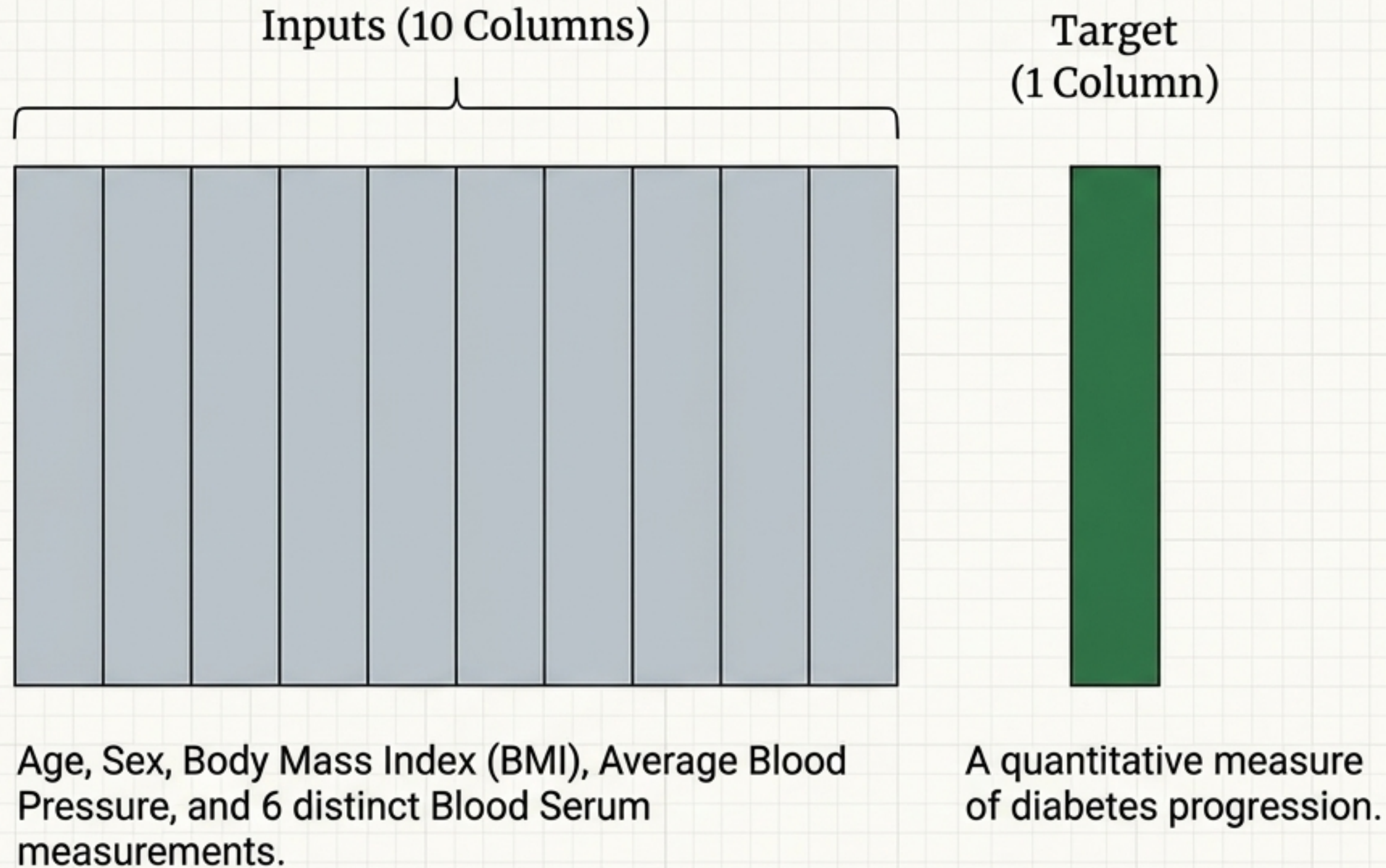


Line 2 (Constrained)



Because $2.03 < 2.25$, the machine learning model chooses the second line. The penalty successfully reduces the slope, resulting in a less complex model.

We can test this constraint practically using the Diabetes Dataset.



Scikit-learn uses the **alpha** hyperparameter to dial the penalty intensity.

```
from sklearn.linear_model import Ridge  
  
model = Ridge(alpha=10)
```

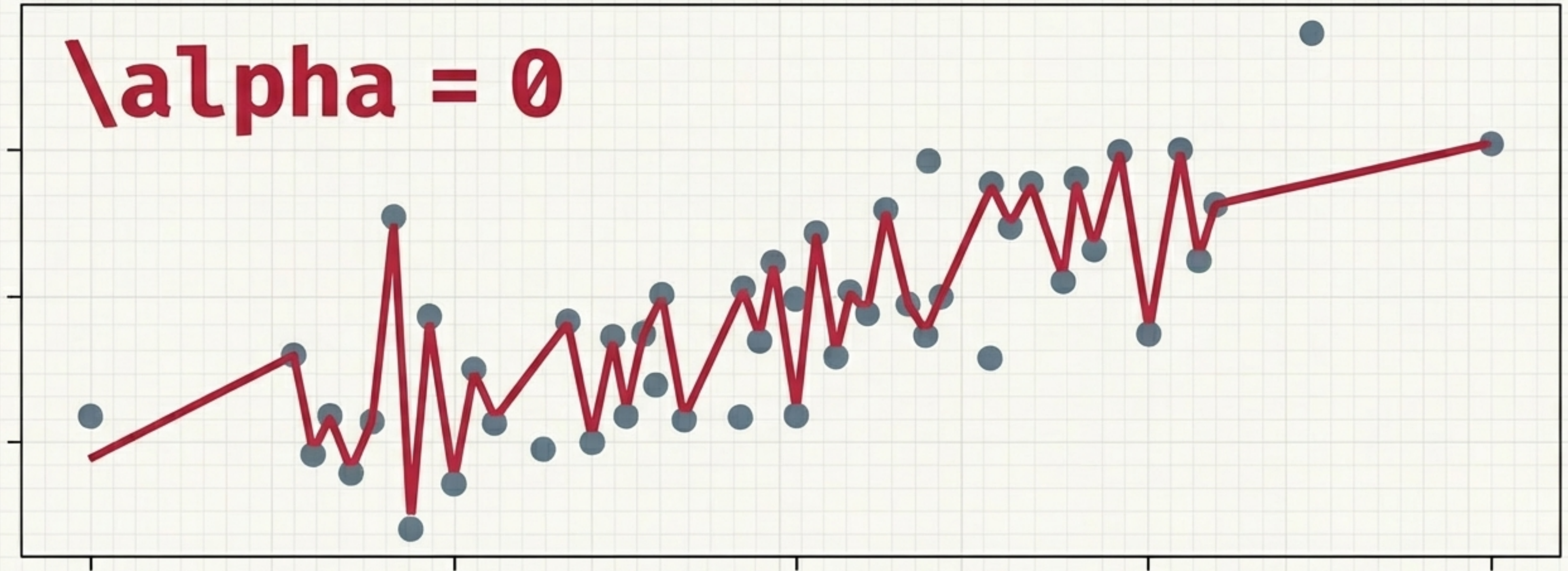


Alpha (α)

Equivalent to theoretical λ

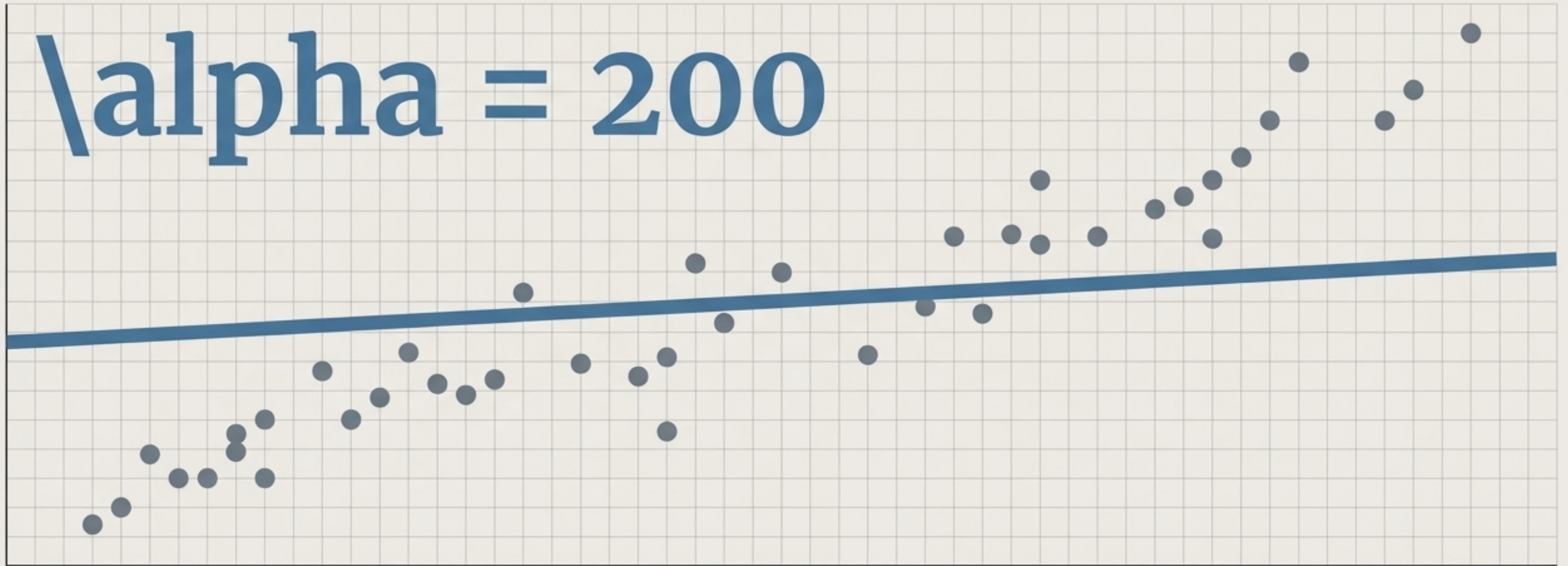
- In practice, **α** acts as the master control dial for the L2 penalty.
- Tuning this hyperparameter determines exactly how strictly we punish model complexity.

Setting **α** to zero disables the penalty and allows overfitting.



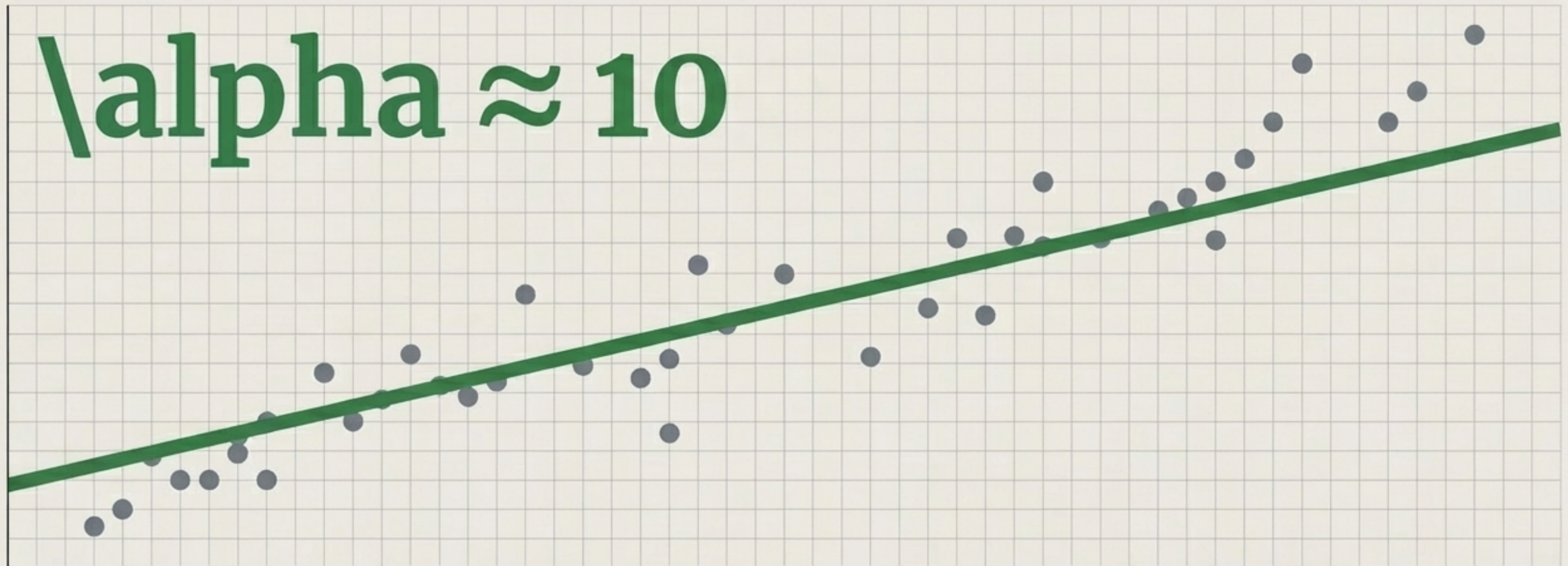
- With no penalty applied, Ridge Regression behaves exactly like standard Linear Regression.
- The red line clearly demonstrates overfitting on the dataset.

An excessively high alpha forces the model to over-simplify and underfit.



- When the alpha value is too high, the model is crushed by the penalty.
- It over-simplifies, fails to pick up basic patterns, and results in underfitting.

An optimal **alpha** balances pattern recognition with structural simplicity.



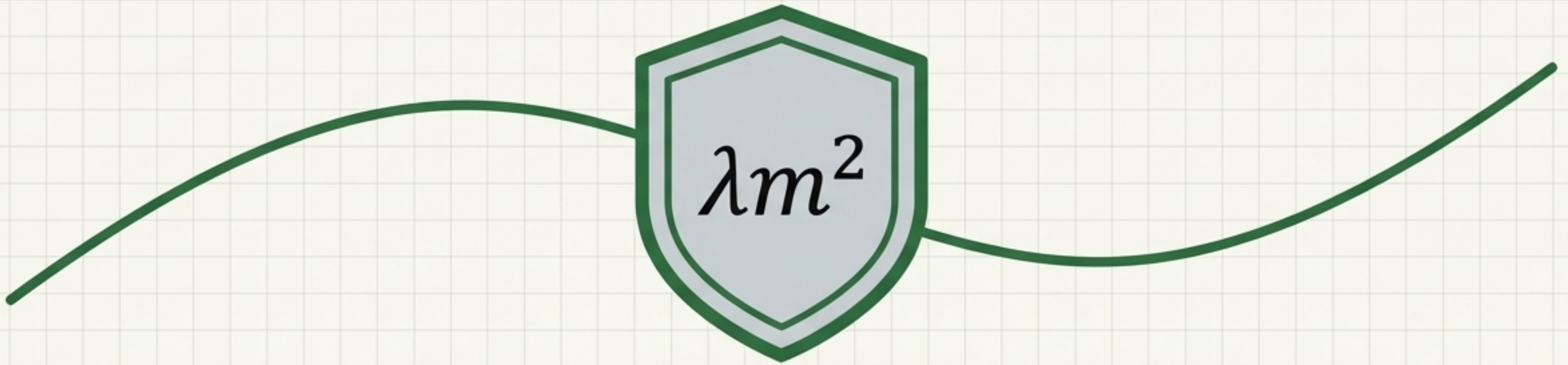
- The optimal alpha achieves the perfect balance.
- The model captures the true underlying pattern without memorising the noise.

Ridge regression delivers comparable predictive accuracy with far less structural complexity.

Standard Linear Regression	Ridge Regression
0.5189 Accuracy Score 	0.5189 Accuracy Score  (Slightly better on newer datasets)

- While the raw accuracy scores on the test data are nearly identical, the underlying model architectures are fundamentally different.
- Ridge achieves the same results without relying on fragile, extreme slopes.

The ultimate goal is generalisation, not memorisation.



- By deliberately adding mathematical bias (λm^2), we reduce model variance.
- Ridge Regression ensures that when your model encounters entirely new data, it relies on true patterns rather than overfitted noise.